

INDIAN INSTITUTE OF TECHNOLOGY TIRUPATI

APE-II FINAL REPORT



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ABSTRACT

Ammonia production is a critical industrial challenge, accounting for 1–3% of global energy usage and nearly 3% of global greenhouse gas emissions due to the energy-intensive nature of the Haber–Bosch process. Given that 85% of ammonia is utilized in fertilizers and its high toxicity complicates transport, localized production has become a vital strategic objective. This study explores the conceptual design of a 50 metric-ton/day plant capable of producing 99.5% purity commercial-grade anhydrous ammonia based on established Linde process guidelines. The design utilizes on-site nitrogen generation via air separation and hydrogen production through steam methane reforming (SMR) of natural gas, followed by water–gas shift conversion and purification to eliminate CO₂ and trace by-products. By integrating material-balance calculations, energy-requirement estimations, and detailed fluid mechanics considerations, this work provides a comprehensive framework for plant design while explicitly addressing the underlying calculation methods, assumptions, and limitations.

1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Ammonia is one of the most widely produced industrial chemicals, primarily due to its central role in fertilizers. Global ammonia production contributes significantly to resource consumption, accounting for 1–3% of world energy use, 5% of natural gas consumption, and nearly 3% of global greenhouse gas emissions. The dominant manufacturing route, the Haber–Bosch process, has historically depended on inexpensive natural gas and remains energy-intensive despite technological advancements. Beyond its industrial relevance, ammonia is hazardous and toxic at ambient conditions, resulting in high transportation costs, strict

regulations, and elevated insurance burdens. Consequently, decentralized or smaller-scale ammonia production facilities can reduce logistics risks and improve accessibility in regions where transport is unsafe or costly.

1.2 PROBLEM STATEMENT

The goal of this project is to design a 50 metric-ton/day ammonia production plant capable of supplying commercial-grade anhydrous ammonia ($\approx 99.5\%$ purity). The design must incorporate safe, reliable, and efficient production of nitrogen and hydrogen feedstocks on-site and achieve the required throughput while minimizing impurities.

1.3 PROCESS OVERVIEW

Ammonia synthesis follows the established Haber–Bosch process, and this study takes guidance from Linde’s ammonia process, known for its industrial reliability.

Key process blocks include:

- Hydrogen Production: Natural gas feed purification: Removal of sulphur compounds (e.g., H_2S) from natural gas followed by Steam methane reforming (SMR) followed by water–gas shift and CO_2 removal.
- Nitrogen production: Using cryogenic distillation, membrane separation, or PSA depending on scale
- Ammonia synthesis loop with recycle and condensation

1.4 FLOW SHEETS MATERIAL BALANCE

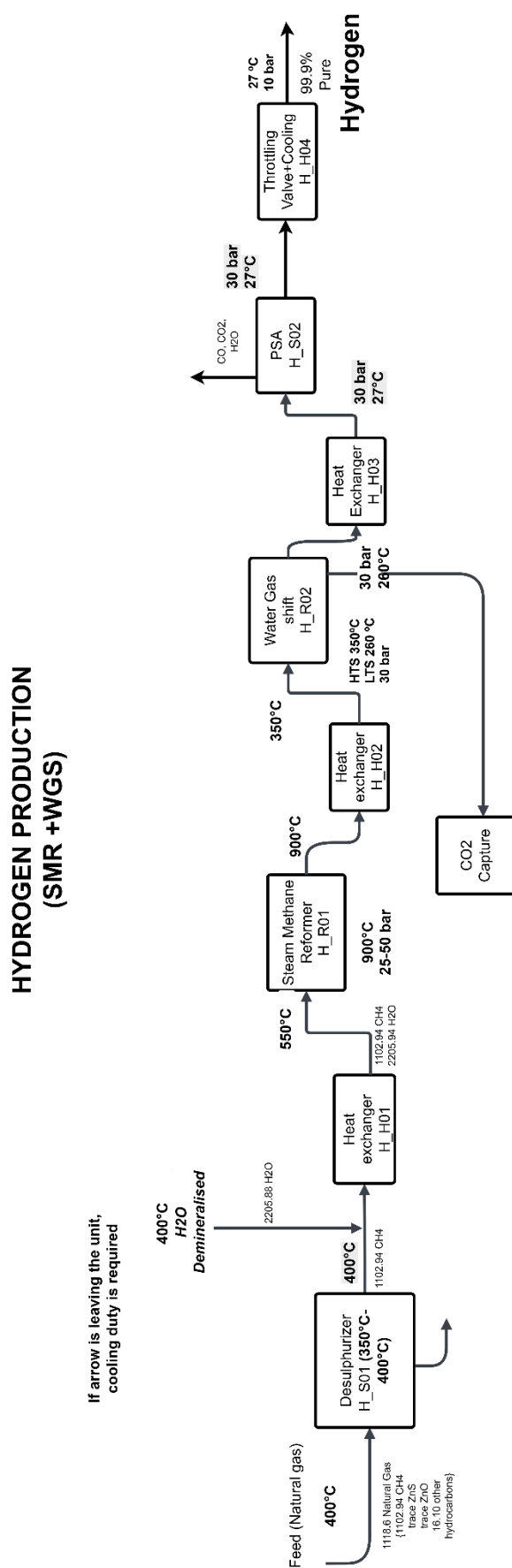


Fig. 1.4.1 Hydrogen Production Material Balance Flow Sheet

ASU (Air Separation Unit) Energy Balance (N2 production)

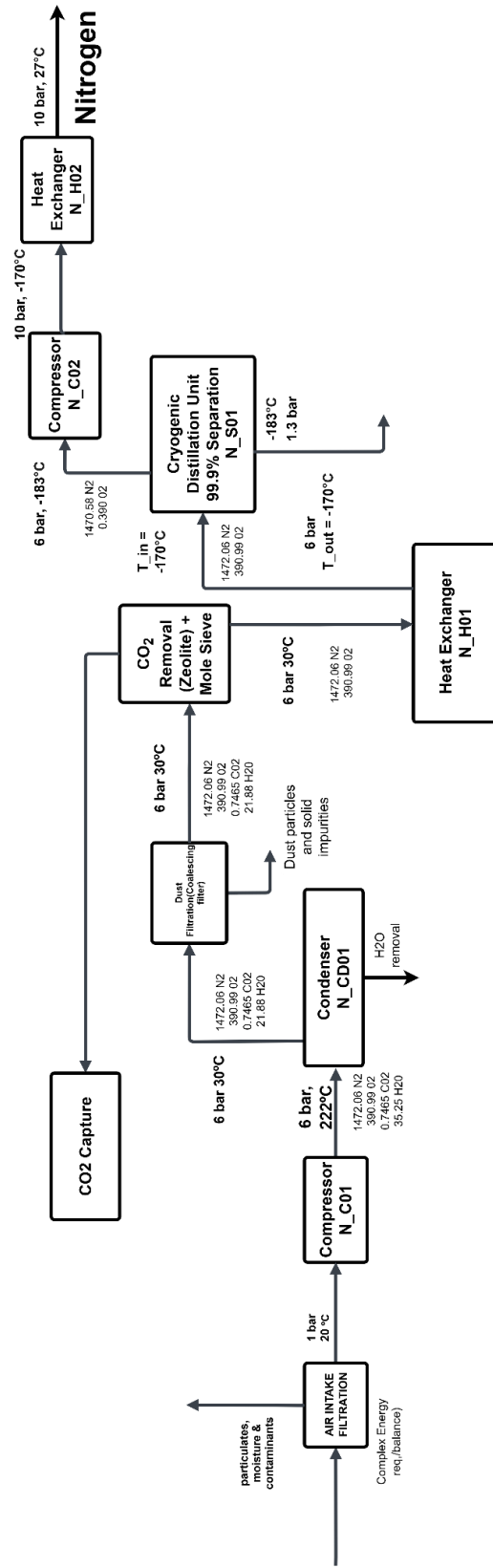
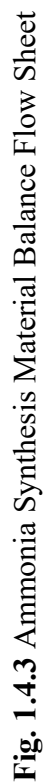


Fig. 1.4.2 Nitrogen Production Material Balance Flow Sheet

All units in kmol/day



1.5 SCOPE

This work presents a comprehensive material balance for the production of 50 tonnes per day of ammonia, with detailed quantification of all input and output streams. The study includes the estimation of raw material requirements, taking into account process stoichiometry and operating conditions and possible inefficiencies in separation or reactions.

In addition, detailed heat balance calculations are performed to evaluate the thermal requirements of the system, including heat consumption and cooling duties across different process units.

Further, this work includes the sizing of pipelines based on calculated flow rates and allowable velocities. Pipe diameters are determined using fluid flow principles, and pressure drops across the piping network are estimated using appropriate correlations and friction factor relationships.

Calculations have been presented in the Supplementary Material attached with this report.

1.6 DISCUSSION

1.6.1 Why Linde's ASU is chosen

1. LAC is a modern modular ammonia production route (Haber-Bosch with optimized front-end).
2. High Purity & Flexibility
3. Scalability
4. PSA (Pressure Swing Adsorption) and membranes are more economical for small plants. But for medium - mega scale, cryogenic (especially Linde's) wins both economically and technically.

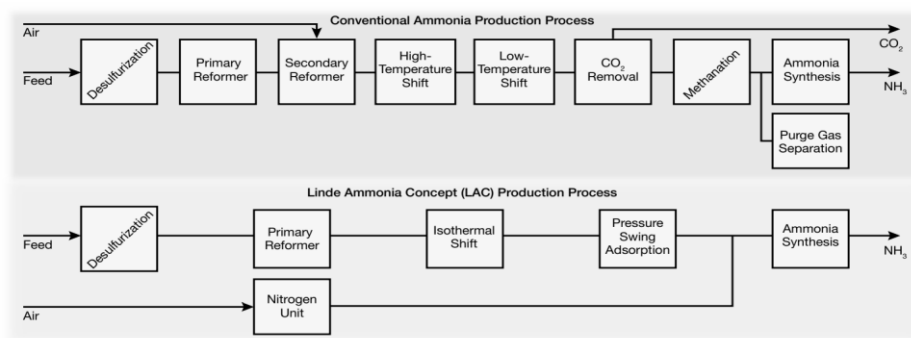


Fig. 1.6.1 Comparison of LAC and conventional Ammonia Production Processes ^[3]

1.6.2 Justification for Cryogenic Separation

Table 1.6.1. Technology comparison table ^[4]

Process	Status	Economic Range (sTPD)	Byproduct capability	Purity Limit (vol.%)	Start-up time
Adsorption	semi-mature	<150	poor	95	minutes
Chemical	developing	undetermined	poor	99+	hours
Cryogenic	mature	>20	excellent	99+	hours
Membrane	semi-mature	<20	poor	40	minutes
ITM	developing	undetermined	poor	99+	hours

1.6.3 Rationale for Continued Use of Natural Gas

- Natural gas is used because technology is already mature and widely available.
- Green hydrogen is not fully commercial yet due to limited electrolyzer capacity and renewable power
- Carbon capture allows continued natural-gas use while reducing emissions.

2. METHODS

2.1 ASSUMPTIONS AND ANALYSIS

1. Ideal gas assumption has been made to calculate the heats of reactions and sensible heats throughout the process.

Heat capacity as a function of temperature was calculated using two empirical correlations. For ideal gas heat capacities, the polynomial form given by Smith et al.^[1] was used:

$$C_p = R(A + BT + CT^2 + D/T^2) \quad \dots (1)$$

where R is the universal gas constant (8.314 J·mol⁻¹·K⁻¹), T is absolute temperature in Kelvin, and A, B, C, D are substance-specific empirical coefficients. Additionally, the Shomate equation from the NIST Chemistry WebBook^[2] was applied:

$$C_p^\circ = A + Bt + Ct^2 + Dt^3 + \frac{E}{t^2} \quad \dots (2)$$

where t = T(K)/1000 and A–E are tabulated Shomate coefficients.

2. For fluid mechanics calculations, Peng Robinson's Equation of State was employed to calculate the densities.

$$P = \frac{RT}{V-b} - \frac{a(T)}{V(V+b)+b(V-b)} \quad \dots (3)$$

$$a(T) = 0.45724 \frac{R^2 T_c^2}{P_c} \alpha(T) \quad \dots (4)$$

$$b = 0.07780 \frac{RT_c}{P_c}$$

$$\alpha(T) = \left[1 + \kappa \left(1 - \sqrt{\frac{T_c}{T}} \right) \right]^2$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$$

3. Compression duty was calculated using the assumption of adiabatic isentropic compression.

$$\dot{W}_s = \dot{n} \times \frac{\gamma}{\gamma-1} \times RT_1 \left[\left(\frac{P_1}{P_2} \right)^{\frac{\gamma}{\gamma-1}} - 1 \right] \quad \dots (5)$$

4. In fluid mechanics calculations, we take a pipe of roughness of 4.5e-5 the roughness of commercial steel. The pipe loss coefficient has been taken as $K_{\text{total}} = 0.04 + 1.00$ (rounded entrance loss + pipe exit loss) and a pipe of length 10 m, no bends has been assumed.

5. Darcy-Weisbach equation was used to calculate pressure drops.

$$\Delta P = f_D \times \frac{L}{D} \times \frac{v^2}{2g} \quad \dots (6)$$

For laminar flow ($Re < 2300$):

$$f_D = \frac{64}{Re} \quad \dots (7)$$

For turbulent flow ($Re > 4000$) — Colebrook equation:

$$\frac{1}{\sqrt{f_D}} = -2 \log \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f_D}} \right) \quad \dots (8)$$

where

ΔP = pressure drop (Pa)

f_D = Darcy friction factor

L = pipe length (m)

D = pipe internal diameter (m)

ρ = fluid density (kg/m³)

v = mean flow velocity (m/s)

g = gravitational acceleration, 9.81 (m/s²)

ε = absolute pipe wall roughness (m)

ε/D = relative roughness

Re = Reynolds number

6. Van der Waals mixing rules have been assumed to be valid.

$$a_{mix} = \sum_i \sum_j y_i y_j a_{ij} \quad \dots (9)$$

where the combining rule is:

$$a_{ij} = \sqrt{a_i a_j} (1 - k_{ij}) \quad \dots (10)$$

y_i, y_j = mole fractions of components i and j

a_i, a_j = pure component attraction parameters

k_{ij} = binary interaction parameter (fitted to experimental data)

$k_{ij} = k_{ji}$ and $k_{ii} = 0$

7. Pressures and temperatures of streams have been taken according to standard industrial practices and in some cases, they are the results of energy balances.

8. i. Efficiency of separator N_S01 is assumed to be 99.9

ii. The reactions of SMR and WGS are assumed to proceed to completion.

iii. Overall conversion of the Ammonia Reactor is assumed to be 20%, reflecting industrial conversion values.

iv. Separation of product and recycle stream is assumed to be 99.9% and minimizes the ammonia vapours passing into recycle stream.

9. Antoine equation has been used to calculate vapour pressures at given temperatures using appropriate coefficients.

2.2 NAMING CONVENTION

Table 2.2.1

Legend	Unit
C	Compressor
CD	Condensor
H	Heat Exchanger
R	Reactor
P	Pump
S	Separator
ST	Stream

2.2 HYDROGEN PRODUCTION SUMMARY

The hydrogen production section begins with desulphurization, followed by preheating, reforming, shift conversion, and gas conditioning. Flow behaviour across the system is primarily influenced by high temperatures and the progressively increasing hydrogen composition, which reduces gas density and increases volumetric flow rates. Pipe sizing is therefore adjusted to maintain acceptable velocities (typically 20–30 m/s) and limit pressure drops, especially as temperatures rise through heat exchangers and peak in the steam methane reformer (approximately 900°C). Hydrogen-rich streams downstream are handled at relatively higher velocities due to their low density. Higher velocity has been prioritised in the calculation of the

From an energy perspective, the process is dominated by the strongly endothermic steam methane reforming reaction, which represents the primary heat demand and governs furnace

duty. This is preceded by significant preheating requirements and followed by the exothermic water-gas shift reaction, whose released heat is recovered through heat exchangers. Downstream cooling and conditioning involve minimal enthalpy changes but support process integration. Overall, the hydrogen section balances high heat input in the reformer with effective heat recovery, while fluid mechanics design ensures efficient transport under low-density, high-temperature conditions.

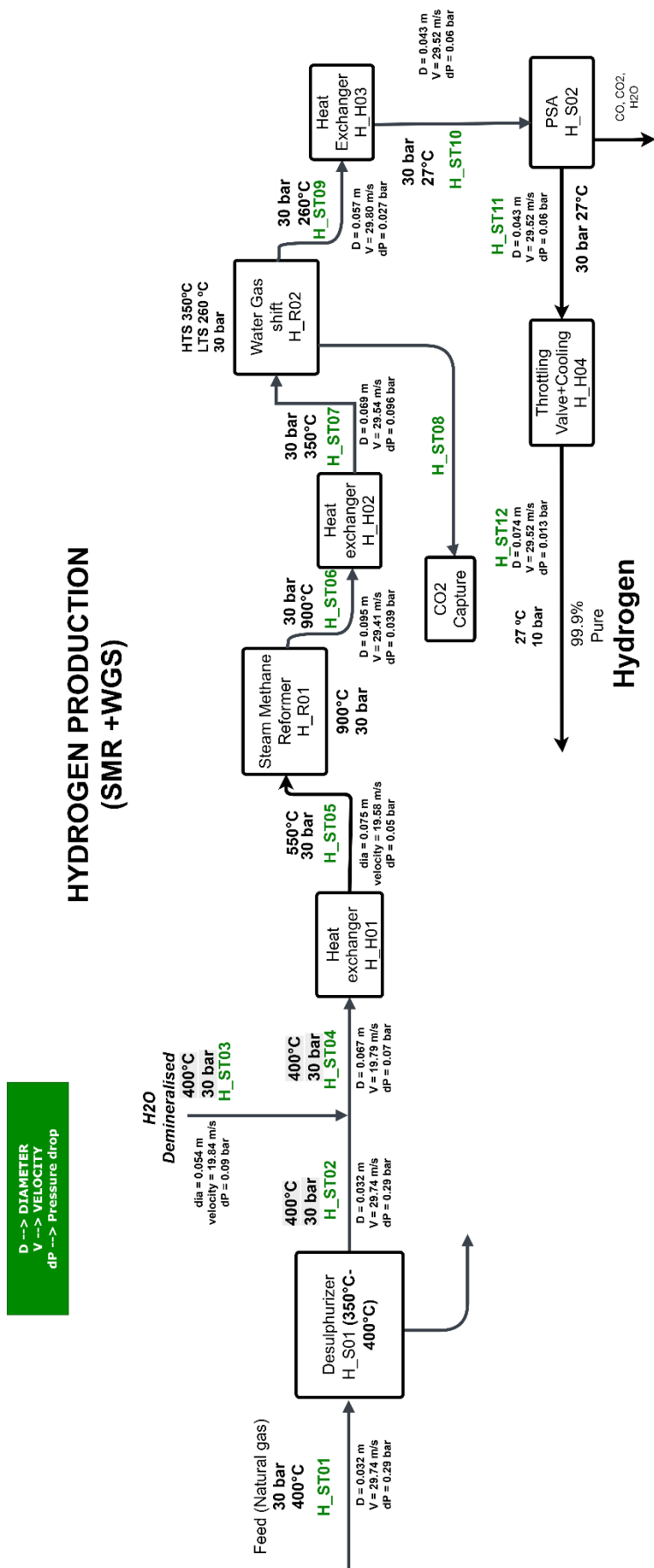


Fig. 2.2.1 Hydrogen Fluid Mechanics Flow Sheet

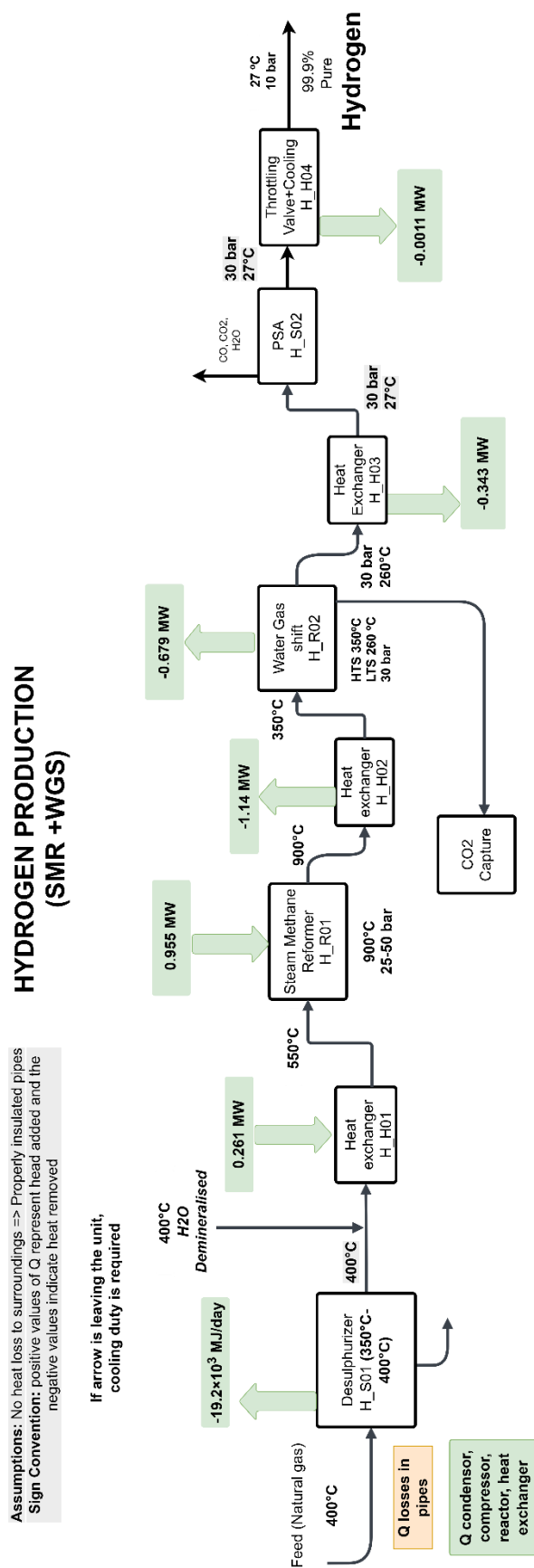


Fig. 2.2.2 Hydrogen Production Heat Duty Flow Sheet

2.3 NITROGEN PRODUCTION (ASU) SUMMARY

The nitrogen section involves air compression, purification, cryogenic separation, and subsequent reheating and recompression, with fluid behaviour strongly influenced by large temperature variations. Initially, compressed air at moderate temperatures and relatively high density is handled at controlled velocities (about 20 m/s) with minimal pressure drops. As the stream is cooled and dried, density increases, allowing more compact piping. In the cryogenic section (approximately at -170°C), the significantly higher density reduces volumetric flow rates, enabling smaller pipe diameters while maintaining stable velocities. Downstream, reheating and recompression restore the gas to ambient conditions, with inert nitrogen permitting relatively higher velocity limits.

Energy-wise, the process is dominated by compression work and cryogenic refrigeration. Initial compression raises enthalpy substantially, followed by cooling and condensation to remove moisture (also an energy intensive step). Energy demand also arises in deep cryogenic cooling, which is essential for effective air separation. Final recompression adds the necessary work to deliver nitrogen at required conditions. Overall, the system balances high energy input for cooling and compression with effective heat integration and stable low-temperature operation.

ASU (Air Separation Unit) Energy Balance (N2 production)

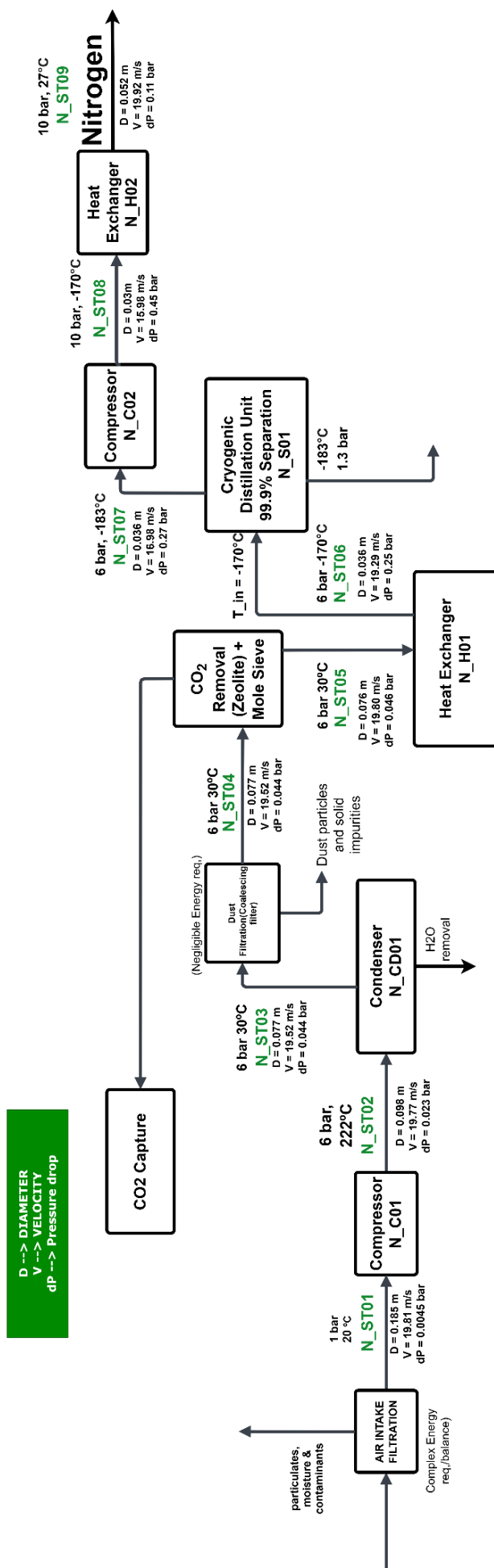


Fig. 2.3.1 Nitrogen Fluid Mechanics Flow Sheet

ASU (Air Separation Unit) Energy Balance (N2 production)

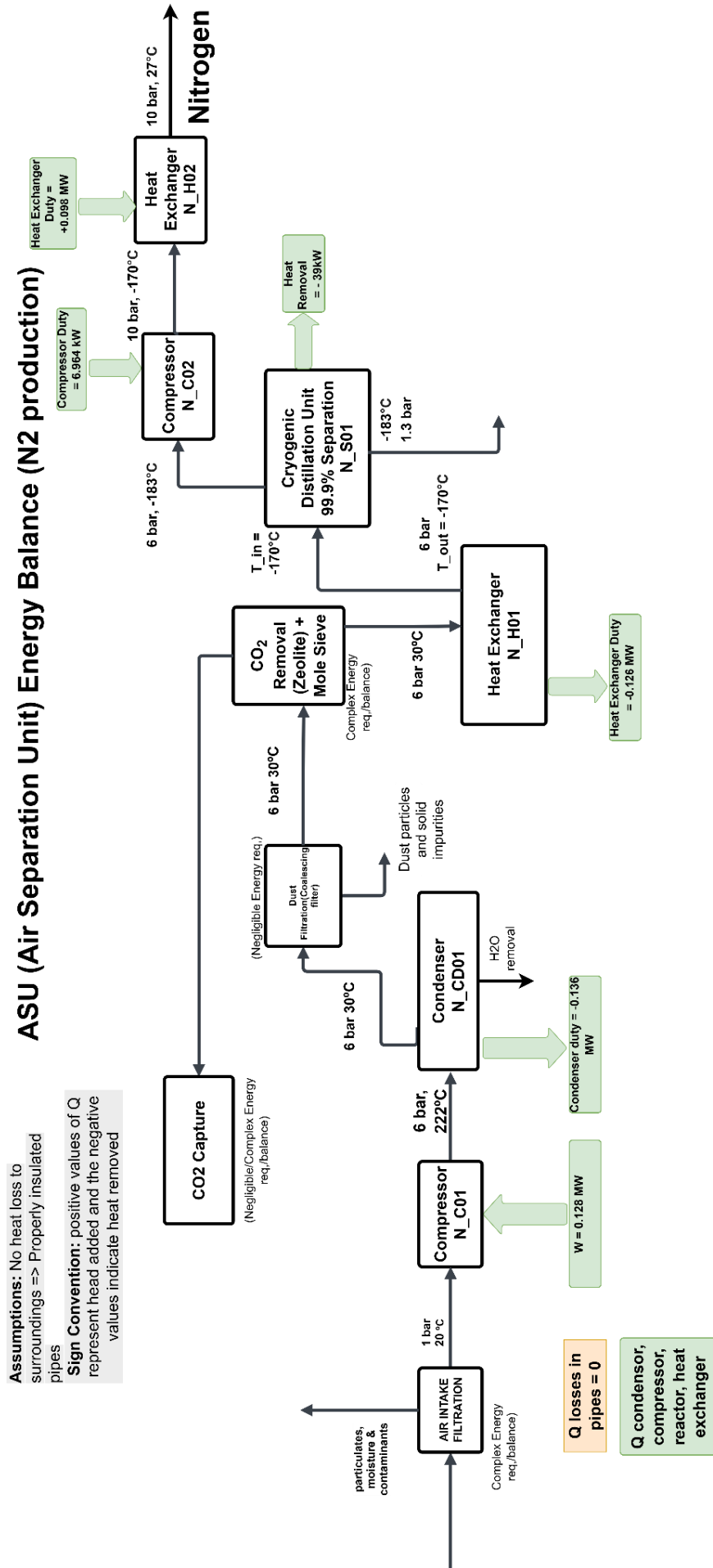


Fig. 2.3.1 Nitrogen Production Heat Duty Flow Sheet

2.4 AMMONIA

The ammonia synthesis loop operates under extreme pressures (of about 200–210 bar), which dominate both fluid flow behaviour and energy requirements. High-pressure compression of feed gases significantly increases density, allowing lower flow velocities (approx. 15 m/s) to minimize pressure drops. Post-reaction throttling reduces pressure (200 to 80 bar).

At 200 bar 450 °C, ammonia is a supercritical fluid and cannot be separated into liquid and gas states. It is throttled to 80 bar where Antoine equation becomes valid again, and then cooled upto -86.3 °C.

The temperature -86.3 °C is chosen so as to satisfy the Antoine equation with very low vapour pressure (0.01 bar).

In the separator, liquid ammonia is separated from unreacted gases, after which recycle streams are recompressed and reheated to maintain loop efficiency. Hydrogen-rich recycle streams permit slightly higher velocities, while high-pressure sections remain conservatively designed.

Energetically, the loop is driven by significant compression work and the exothermic nature of ammonia synthesis. Compression stages contribute the largest energy input, with temperature increase. The synthesis reaction releases substantial heat, requiring continuous removal to maintain near-isothermal reactor conditions, with opportunities for heat recovery. Throttling provides cooling with negligible enthalpy change, aiding downstream condensation. The condenser represents a major heat removal step essential for ammonia recovery and recycle control. Overall, the process balances high compression energy input with effective heat removal, recovery, and recycling to achieve efficient conversion.

HABER BOSCH PROCESS AMMONIA SYNTHESIS

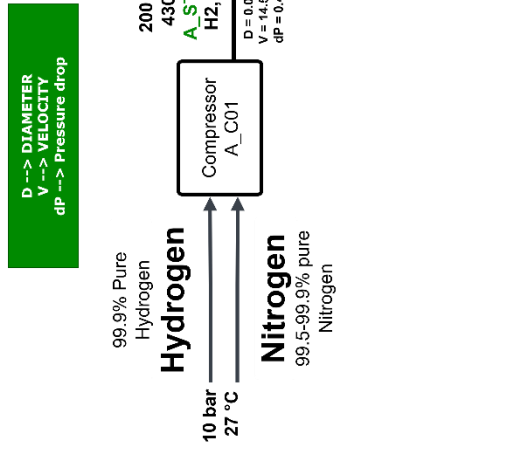


Fig. 2.4.1 Ammonia Synthesis Fluid Mechanics Flow Sheet

HABER BOSCH PROCESS AMMONIA SYNTHESIS

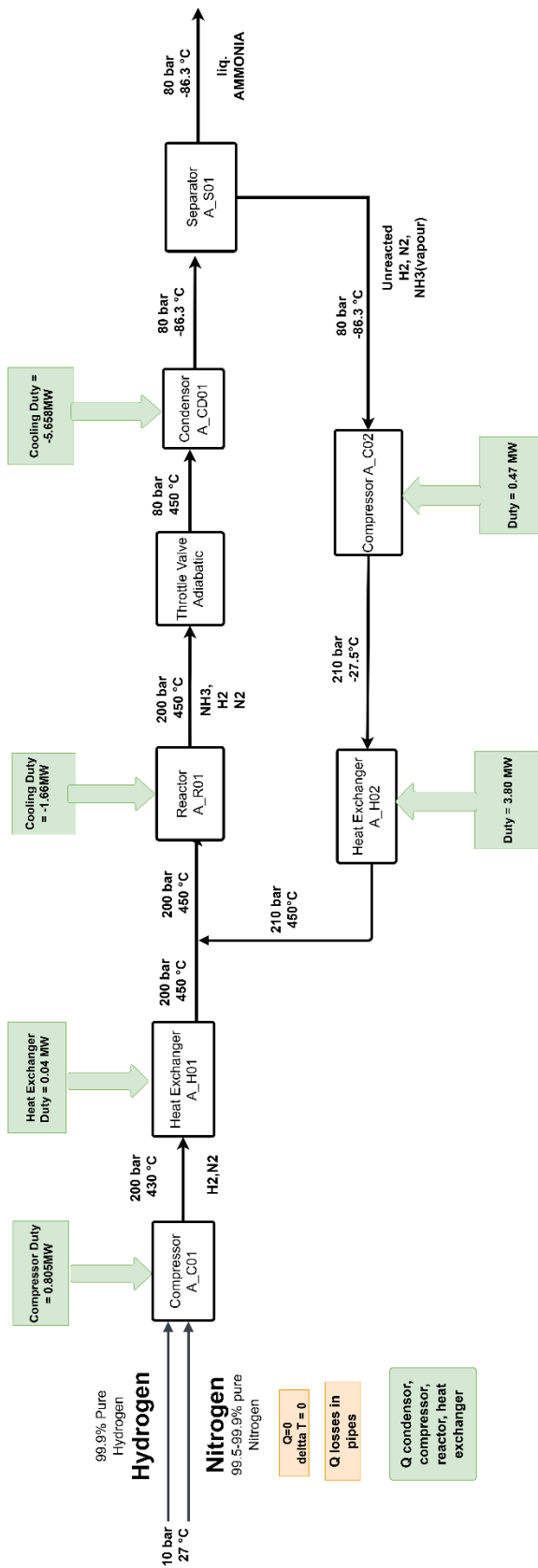


Fig. 2.4.2 Ammonia Synthesis Heat Duty Flow Sheet

3. RESULTS AND DISCUSSION

3.1 NUMERICAL DATA

Overall Plant Material Balance

The integrated ammonia production plant is designed to produce 50 tonnes of liquid ammonia per day. To sustain this output, the plant requires a **total air intake of 1899 kmol/day** fed to the Air Separation Unit, a **natural gas feed of 1118.6 kmol/day** and **demineralised water of 2206 kmol/day** supplied to the hydrogen production unit. These three feed streams collectively drive the entire upstream production of hydrogen and nitrogen, which are subsequently combined in the stoichiometric 3:1 molar ratio within the Haber-Bosch synthesis loop to meet the target ammonia output.

Overall Plant Energy Balance

The total energy demand of the plant is distributed across three major process units. **Hydrogen production** via Steam Methane Reforming and Water Gas Shift constitutes the most energy-intensive upstream operation, requiring 1.128×10^8 kJ/day, driven primarily by the high-temperature endothermic reforming reaction. **Nitrogen production** through the Air Separation Unit consumes 4.94×10^6 kJ/day, predominantly attributed to the compression and cryogenic distillation stages. The **Haber-Bosch ammonia synthesis loop** accounts for an energy requirement of 4.65×10^{11} kJ/day, reflecting the combined duties of feed compression, reactor operation, condensation, and recycle compression at high pressure.

Fluid Mechanics and Pipe Sizing

Across all three process units, pipe diameters were sized to maintain velocities within industrially accepted ranges, ensuring both hydraulic efficiency and structural integrity. In the Haber-Bosch unit, operating at pressures up to 200 bar, pipe diameters range from 0.043 m to

0.1 m with velocities between 14.55 m/s and 29.31 m/s and pressure drops not exceeding 0.73 bar per segment. The hydrogen production pipework, operating at 25–50 bar, employs diameters between 0.032 m and 0.075 m with velocities consistently near 29.5 m/s and segment pressure drops as low as 0.013 bar. The ASU pipework spans the widest diameter range, from 0.03 m at compressed nitrogen delivery to 0.185 m at the atmospheric air intake, reflecting the large density variation from ambient to cryogenic conditions, with velocities uniformly maintained between 15 m/s and 20 m/s and a maximum segment pressure drop of 0.45 bar. Across all units, the calculated pressure drops confirm that the designed pipe network operates within acceptable hydraulic limits without requiring additional pressure compensation measures.

3.2 RATIONALE ON MAXIMUM ALLOWABLE VELOCITY AND PRESSURE DROPS

Table 3.2.1 Stream Descriptions

Stream	Pressure (bar)	Temperature (°C)	Velocity Max	P Drop Max	Reasoning
N_ST01	1	20	20	0.05	dense, non flammable
N_ST02	6	222	20	0.3	dense, non flammable
N_ST03	6	30	20	0.3	dense, non flammable
N_ST04	6	30	20	0.3	dense, non flammable
N_ST05	6	30	20	0.3	dense, non flammable
N_ST06	6	-170	20	0.3	dense, non flammable
N_ST07	6	-183	20	0.3	dense, non flammable
N_ST08	10	-170	20	0.5	dense, non flammable
N_ST09	10	27	20	0.5	dense, non flammable
A_ST01	200	430	15	2	High P streams

A_ST02	200	450	15	2	High P streams
A_ST03	200	450	15	2	High P streams
A_ST04	200	450	15	2	High P streams
A_ST05	80	450	30	2.4	H2 rich
A_ST06	80	-86.3	30	2.4	H2 rich
A_ST07	-	-	-	-	-
A_ST08	80	-86.3	30	2.4	H2 rich
A_ST09	210	-27.5	15	2.1	High P stream
A_ST10	210	450	15	2.1	High P stream
H_ST01	30	400	30	0.9	fuel gas
H_ST02	30	400	30	0.9	fuel gas
H_ST03	30	400	20	1.5	dense, non flammable
H_ST04	30	400	20	1.5	dense, non flammable
H_ST05	30	550	20	1.5	dense, non flammable
H_ST06	30	900	30	0.9	H2 rich
H_ST07	30	350	30	0.9	H2 rich
H_ST08	-	-	-	-	-
H_ST09	30	260	30	0.9	H2 rich
H_ST10	30	27	30	0.9	H2 rich
H_ST11	30	27	30	0.9	H2 rich
H_ST12	10	27	30	0.3	H2 rich

3.3 LIMITATIONS (AND FURTHER IMPROVEMENTS)

1. The reactors have not been optimized for volume in this work – the industrial practice of 20% conversion has been taken, with product and recycle stream separator efficiency of 99.9%.
2. SMR and WGS reactions may not proceed to completion in reality. PSA is the unit where unreacted species are separated, but it will result in slight increase in feed of natural gas to achieve the 50 tonne/day ammonia production.
3. Heat Exchanger routing may ensure that heat gained in a heat exchanger passes to the heat lost in the other, increasing energy efficiency of the plant. The routing of heat exchangers has

not been included in this report. Also, heat exchanger design and energy of flowing fluid through heat exchangers has not been included.

4. Control systems have not been employed in the present report.

5. Enthalpy and entropy have not been changed considering the mixture equilibrium and non-ideality of solution.

4. CONCLUSION

This study presents a comprehensive conceptual design of a 50 metric-ton per day ammonia production plant based on the Linde ammonia process, integrating hydrogen production via Steam Methane Reforming, nitrogen supply through cryogenic Air Separation, and the Haber-Bosch synthesis loop. The material balance, energy balance, and fluid mechanics calculations across all three process units were successfully completed and found to be mutually consistent with the target production specification.

The upstream feed requirements determined from the material balance align well with the stoichiometric demands of the synthesis loop, confirming the internal consistency of the overall plant design. Among the three process units, hydrogen production was identified as the dominant energy consumer, driven by the strongly endothermic reforming reaction, while the ASU operates with comparatively modest energy demands owing to effective heat integration across its compression and cryogenic stages. The Haber-Bosch loop, while the most mechanically intensive due to its extreme operating pressures, benefits from significant heat recovery potential within the recycle circuit.

The fluid mechanics analysis confirmed that all process pipework was successfully sized within industrially accepted velocity and pressure drop limits, with design choices appropriately

reflecting the distinct fluid properties encountered across each unit, from high-pressure supercritical mixtures in the synthesis loop to cryogenic nitrogen streams in the ASU.

Key limitations of this work include the assumption of ideal gas behaviour, fixed reactor conversion at the industrial benchmark of 20%, and the absence of heat exchanger network optimisation and process control design. Addressing these aspects in future work would further improve the accuracy and energy efficiency of the plant design. Overall, this study establishes a rigorous and self-consistent conceptual framework for a modular ammonia production facility, demonstrating the technical feasibility of the integrated process at the target production scale.

5. REFERENCES

- [1] Smith, J.M., Van Ness, H.C., and Abbott, M.M. (2017). *Introduction to Chemical Engineering Thermodynamics*, 8th ed. McGraw-Hill, New York. Appendix C.
- [2] National Institute of Standards and Technology (NIST). *NIST Chemistry WebBook, SRD 69*. Retrieved from <https://webbook.nist.gov>
- [3] Appl, M. (2012). A short history of the ammonia process: Past, present and future. Retrieved from <https://ammoniaknowhow.com/short-history-of-ammonia-process-past-present-and-future/>
- [4] Smith, A.R., and Klosek, J. (2001). *A review of air separation technologies and their integration with energy conversion processes*. Fuel Processing Technology, 70(2), 115–134. Retrieved from <https://netl.doe.gov/sites/default/files/2023-08/Smith%202001%20Review%20of%20Air%20Separation%20Technologies.pdf>

6. SUPPLEMENTARY MATERIAL

6.1 CODE DESCRIPTIONS

S. No.	Section	Process	Code Name	Description	Calculated Quantity
1	Ammonia	Compressor 1 A_C01	AM_01	Adiabatic Isentropic Compression of H ₂ , N ₂ (10 bar to 200 bar, 300K to 713K)	T ₂ , Work and DeltaH
2	Ammonia	Hypo Compressor 1 A_C01	AM_02	Adiabatic Isentropic Compression of H ₂ , N ₂ (10 bar to 200 bar, 300K to 713K), Gamma constant 1.41	Work and DeltaH
3	Ammonia	Heat Exchanger 1 A_H01	AM_03	Heat to Reactor Temperature, taken T ₁ from AM_01	Q sensible (heating)
4	Ammonia	Mixing	NA	Streams Feed and Recycle are mixed at 200 bar 500 deg	NA
5	Ammonia	Reactor A_R01	AM_04	T _{in} , T _{out} , Fresh feed, Recycled feed, Overall Conversion 20%, Heat of Reaction, Sensible Heats, ASSUMED ISOTHERMAL REACTOR 500 DEG CELSIUS OPERATIONS	Q total , T ₂ .
6	Ammonia	Throttler A_C_02	AM_05	Throttling temperature rise or fall approximately 0 400 bar to 80 bar	Condensor Duty
7	Ammonia	Condensor A_CD01	AM_06	Condensor Duty to Cool Ammonia to certain gas - liq ratio (product - recycle ratio)	T ₂ , Work
8	Ammonia	Compressor A_C02	AM_07	To compress N ₂ H ₂ recycled NH ₃ vapors to 210 bar from 80 bar	

9	Ammonia	Heat Exchanger	AM_08	Heat Recycle Stream to 450 deg Celsius for mixing again	Work
10	Nitrogen	Compressor 1 N_C01	N2_01	To compress low pressure air for easier transport	T2, Work and DeltaH
11	Nitrogen	Condensor 1 N_CD01	N2_02	For separating moisture and avoid ice in cryo unit., removes 60% water content, zeolite will remove CO2 and H2O leftover. Can be optimised further for maximum energy saving and purity.	Cooling duty, composition after condensation, temperature final
12	Nitrogen	Heat Exchanger 1 N_H01	N2_03	To bring to cryogenic temperatures i.e. 30 deg Celsius to -170 deg Celsius, used NIST Webbook Cp range 100 to 500K	Power rating
13	Nitrogen	Cryo Unit N_S01	N2_04	Cryogenic Unit Energy Balance	Energy Balance
14	Nitrogen	Heat Exchanger 2 N_H02	N2_05	Post cryogenic temperatures i.e. -170 to 27 deg Celsius, used NIST Webbook Cp range 100 to 500K	
15	Nitrogen	Compressor 2 N_C02	N2_06	Post Cryo Pressure Increase	Compression final temp, compressor duty
16	Hydrogen	Desulphurizer H_S01	H2_01	Calculated using Enthalpy of Desulphurization	Power
17	Hydrogen	Heat Exchanger 1 H_H01	H2_02	CH4 and H2O after Desulph to SMR 400 TO 550 deg C	Power
18	Hydrogen	Heat Exchanger 2 H_H02	H2_03	CH4 , H2O, CO, H2O after SMR to WGS 400 TO 550 deg C	Power
19	Hydrogen	Reactor 1 H_R01	H2_04	SMR Tin, Tout, deltaHrxn	Heating cooling duty

20	Hydrogen	Reactor 2 H_R02	H2_05	WGS Tin, Tout, delta Hrxn	Heating cooling duty
21	Hydrogen	Heat Exchanger H_H03	H2_06	Heat Exchange before PSA	Heating cooling duty
22	Hydrogen	Throttling H_H04	H2_07	Throttling + Cooling After PSA	Heating cooling duty
23	All	All	main	All streams	Stream compositions and PipeDia fxn call
24	All	All	PipeDia	Any stream	Velocity and diameter calculation

6.2 CALCULATIONS

All calculation codes and flowsheets referenced in this report are available at the following link:

[Supplementary Codes – Google Drive](#)